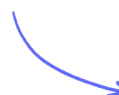


Sets

Set Notation & Venn Diagrams

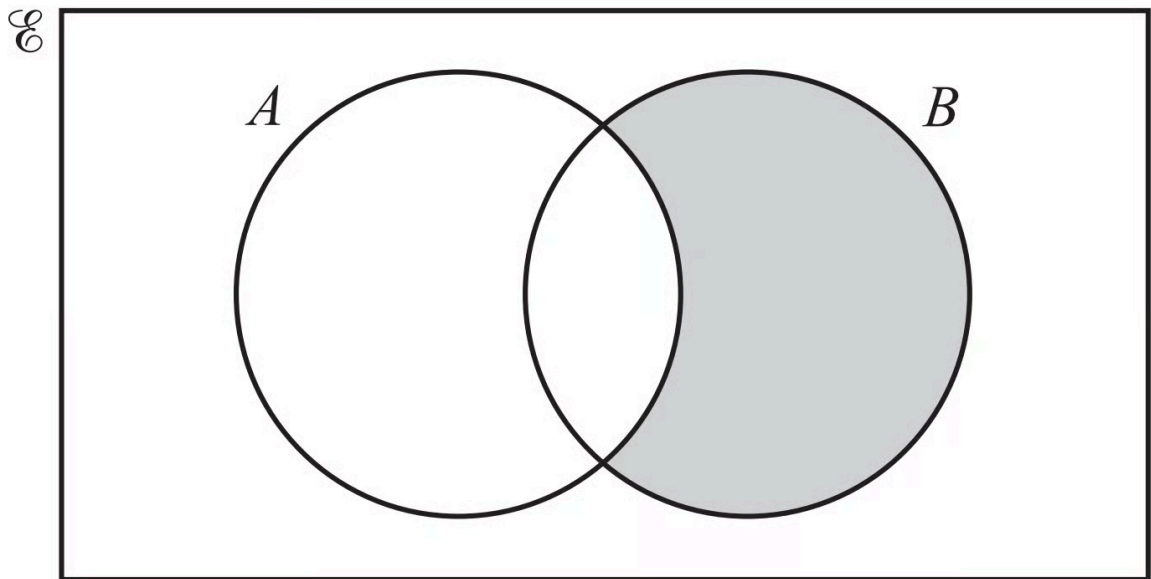
Easy (6 questions)	/7
Medium (7 questions)	/19
Hard (7 questions)	/12
Very Hard (3 questions)	/3
Total Marks	/41

Scan here to return to the course
or visit [savemyexams.com](https://www.savemyexams.com)



Easy Questions

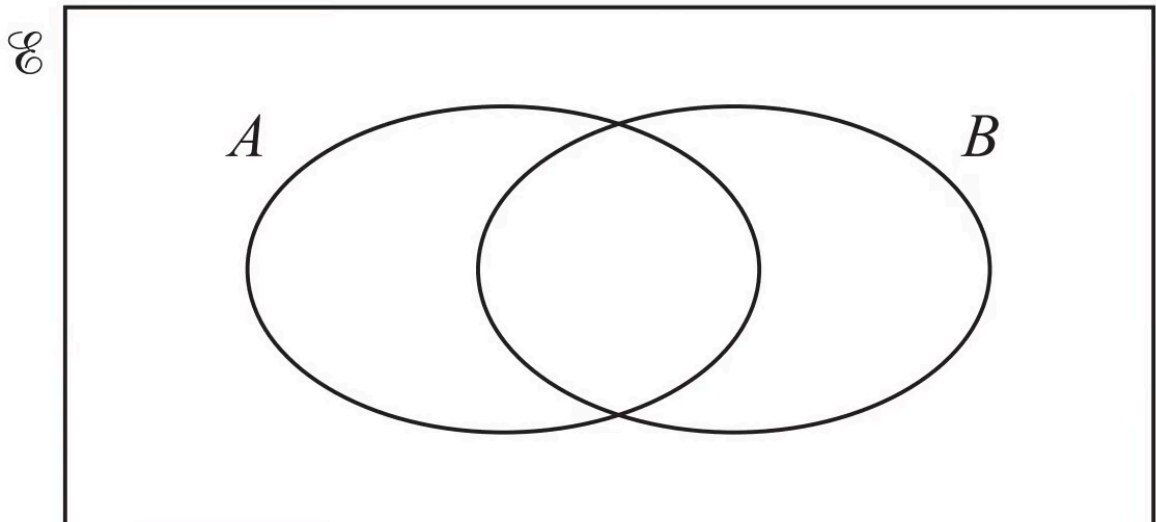
1



Use set notation to describe the shaded region.

(1 mark)

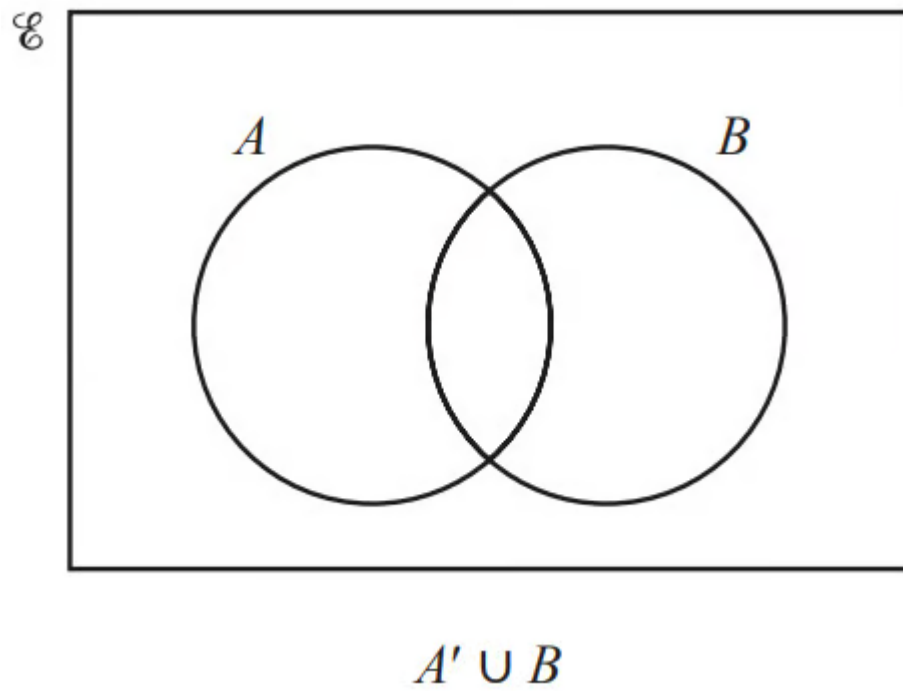
2



On the Venn diagram, shade the region $A \cap B$.

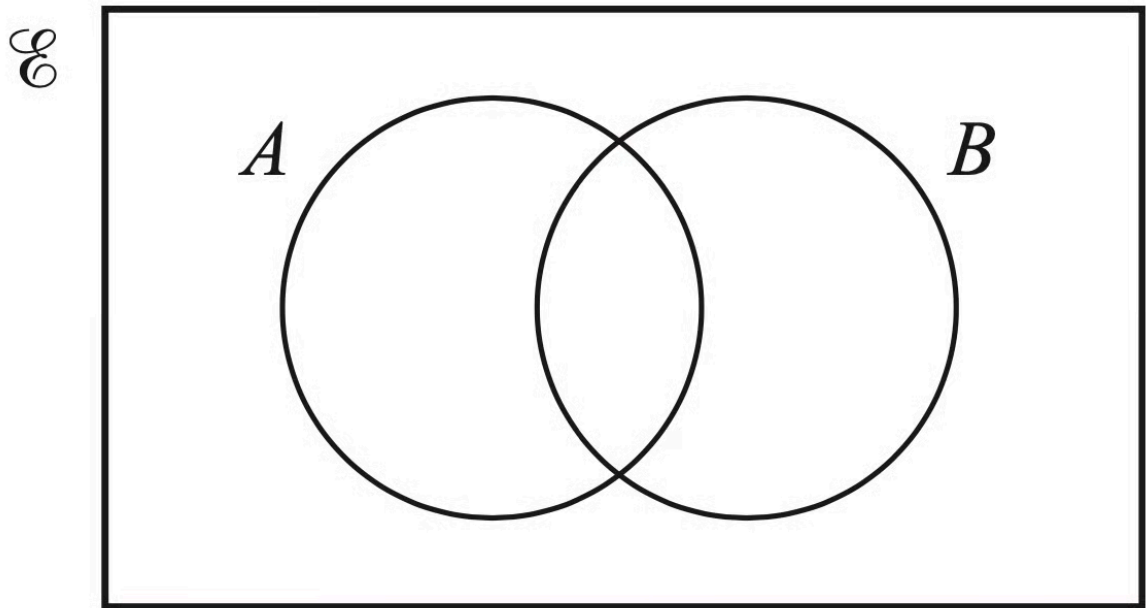
(1 mark)

- 3 Shade the region $A' \cup B$ in the Venn diagram below.



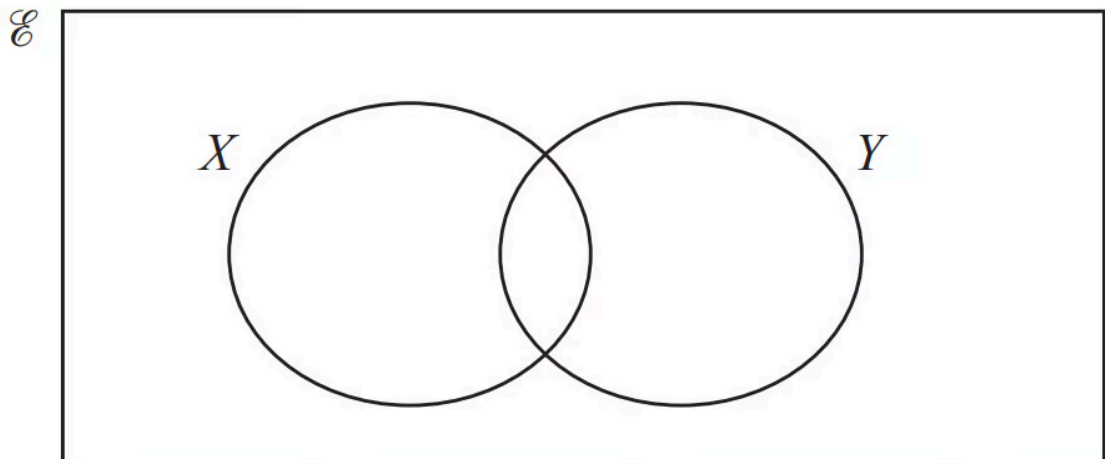
(1 mark)

- 4 On the Venn diagram, shade the region $(A \cap B)'$.



(1 mark)

- 5 In the Venn diagram, shade $X' \cap Y$.



(1 mark)

- 6 $\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$A = \{\text{odd numbers}\}$

$B = \{\text{multiples of 3}\}$

List the members of the set

i) $A \cap B$

[1]

ii) $A \cup B$

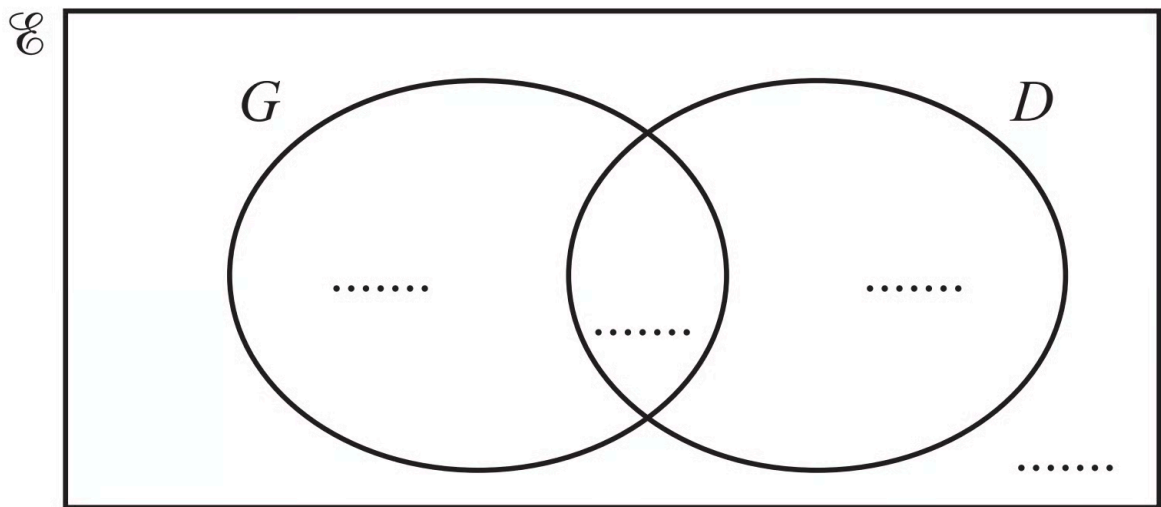
[1]

(2 marks)

Medium Questions

1 In a class of 40 students:

- 28 wear glasses (G)
- 13 have driving lessons (D)
- 4 do not wear glasses and do not have driving lessons.



i) Complete the Venn diagram.

[2]

ii) Use set notation to describe the region that contains a total of 32 students.

[1]

(3 marks)

2 $E = \{0, 1, 2, 3, 4, 5, 6\}$ $A = \{0, 2, 4, 5, 6\}$ $B = \{1, 2, 5\}$

Complete each of the following statements.

$$A \cap B = \{ \dots \}$$

$$n(B) = \dots$$

$$\{0, 4, 6\} = \dots \cap \dots$$

$$\{2, 4\} \dots A$$

(4 marks)

3 $M = \{x: x \text{ is an integer and } 2 \leq x < 6\}$

i) Find $n(M)$.

[1]

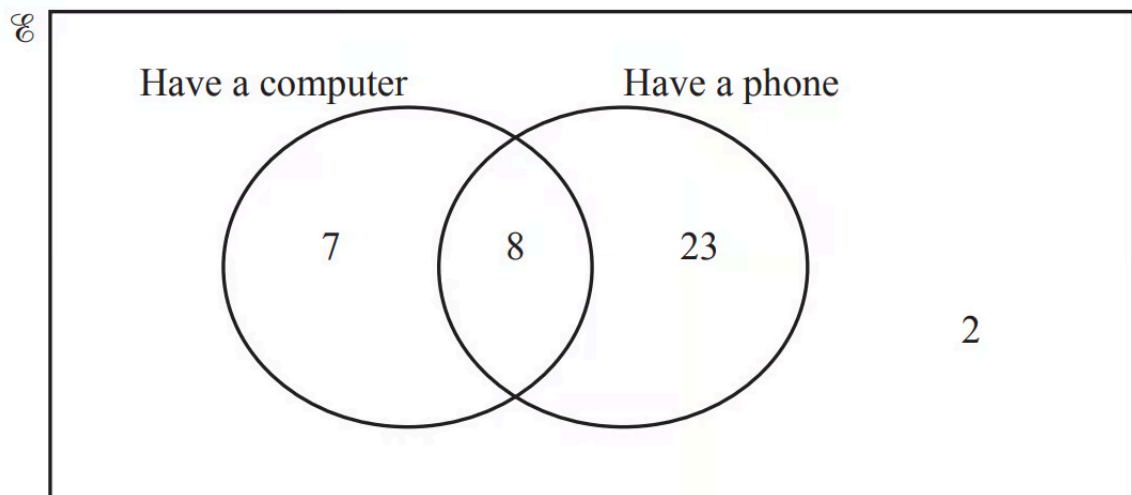
ii) Write down a set N where $N \subseteq M$ and $N \neq \emptyset$.

{.....} [1]

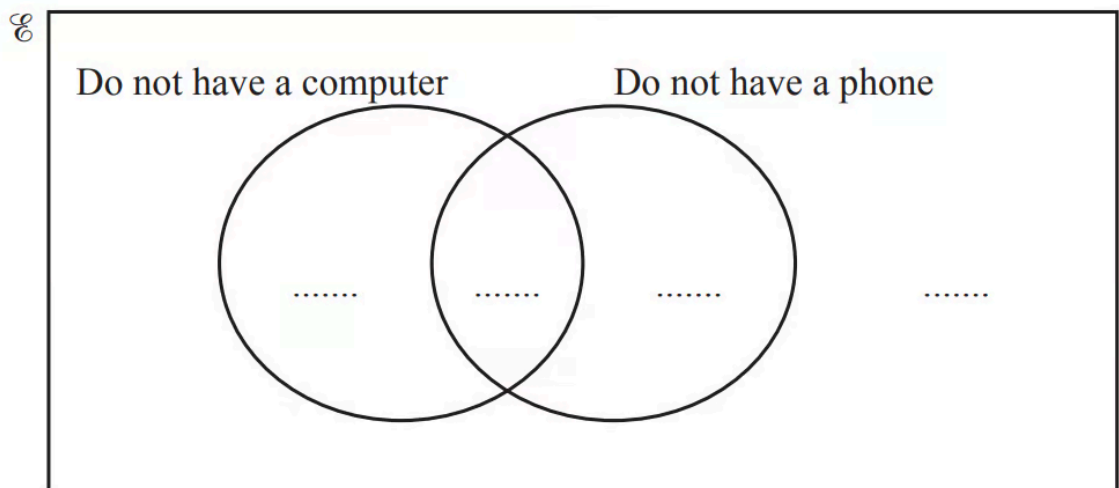
(2 marks)

4 40 children were asked if they have a computer or a phone or both. The Venn diagram

shows the results.



Complete the Venn diagram.



(2 marks)

5 (a) $C = \{x : x \text{ is an integer and } 5 < x < 12\}$ $D = \{5, 10\}$

Select the correct statement from the list below.

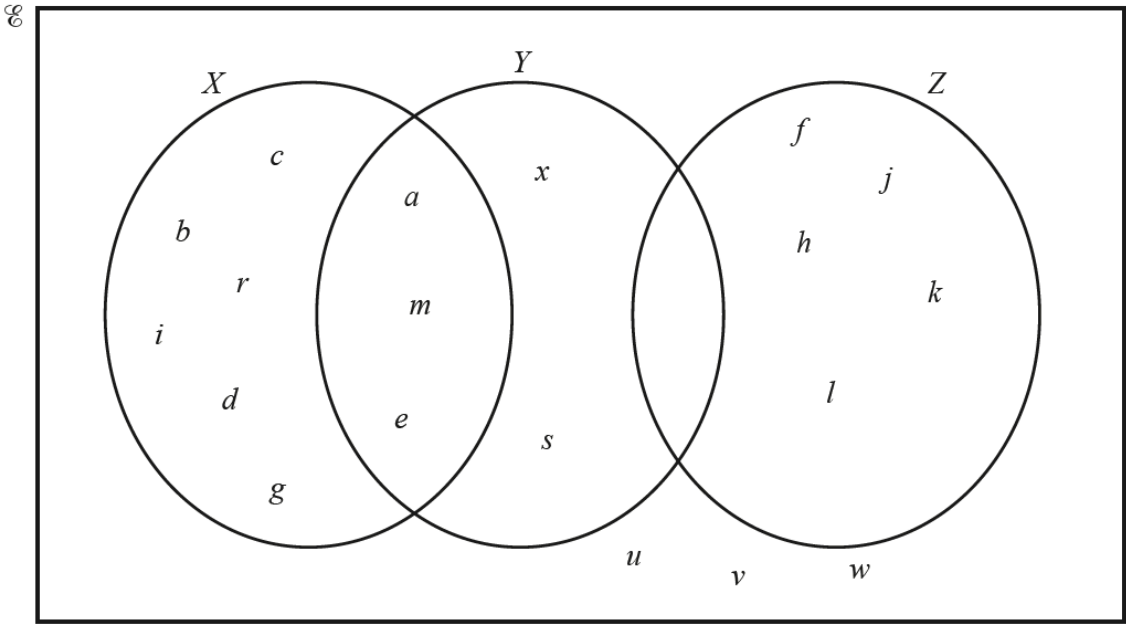
$D = \emptyset$ $C \cap D = \{10\}$ $6 \in D$ $D \subseteq C$

(1 mark)

(b) Find $n(C \cup D)$.

(1 mark)

6



Use set notation to complete the statements for the Venn diagram above.

i) $c \dots\dots\dots X$

[1]

ii) $\dots\dots\dots = \{a, m, e\}$

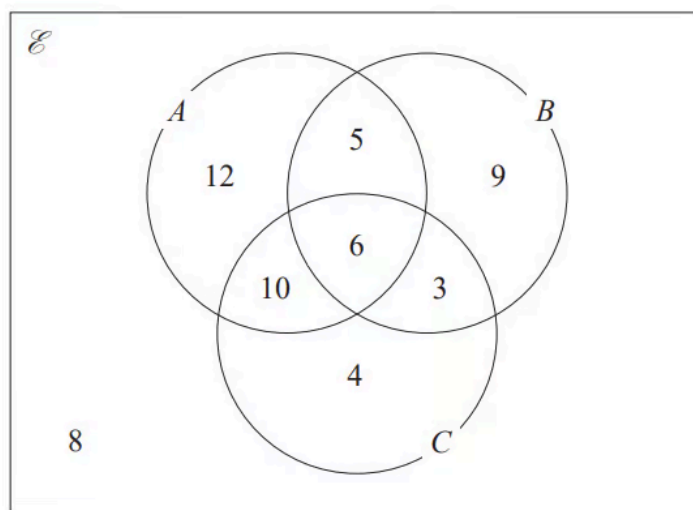
[1]

iii) $Y \cap Z = \dots\dots\dots$

[1]

(3 marks)

7 The Venn diagram shows a universal set, $\mathcal{E}$ and sets A , B and C .



12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

i) $n(A \cup B)$

[1]

ii) $n(A' \cap B')$

[1]

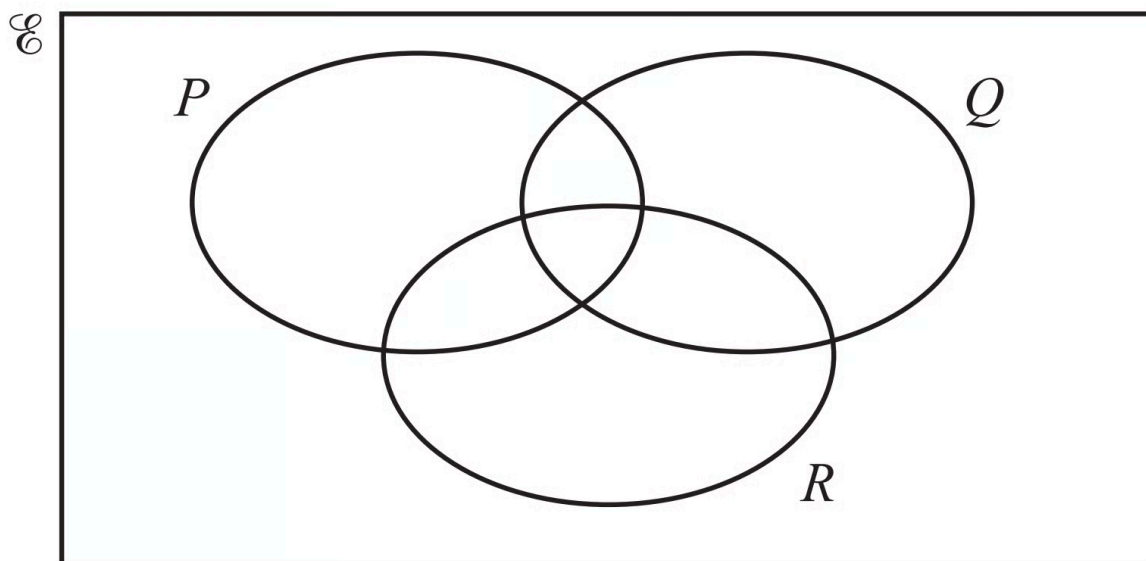
iii) $n([A \cap B] \cup C)$

[1]

(3 marks)

Hard Questions

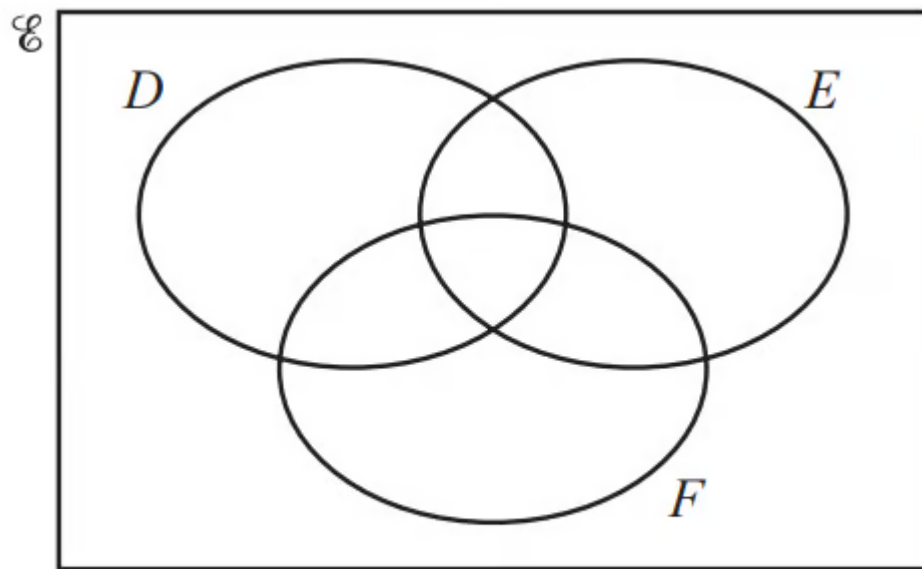
1



Shade the region $P \cup (Q \cap R)'$.

(1 mark)

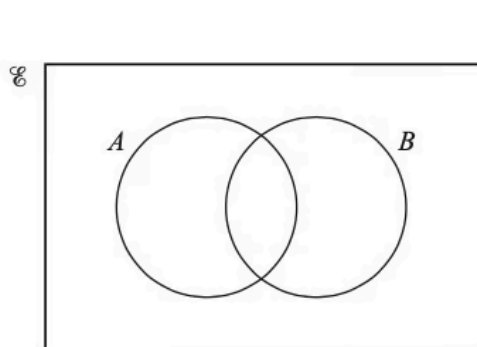
- 2 Shade the region $(D \cap E)' \cap F$ in the Venn diagram below.



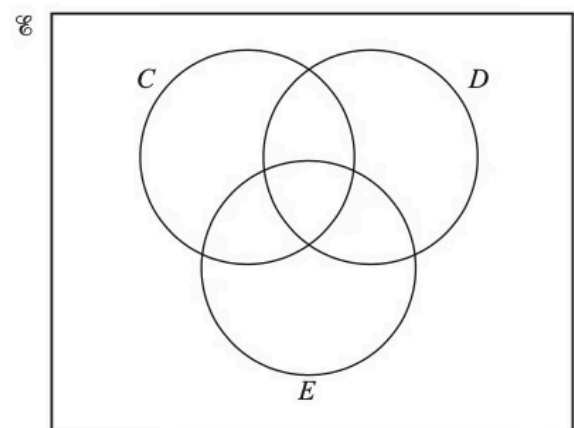
$$(D \cap E)' \cap F.$$

(1 mark)

- 3 In each Venn diagram, shade the required region.



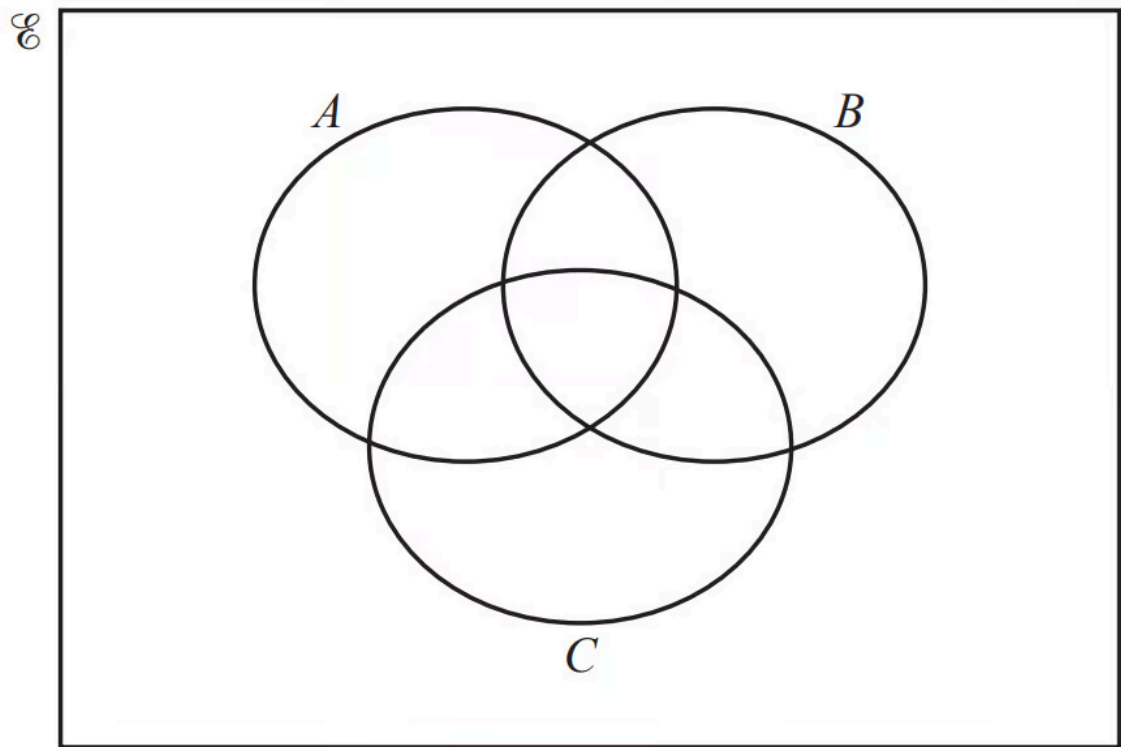
$$(A \cup B)'$$



$$(C \cap D') \cup E$$

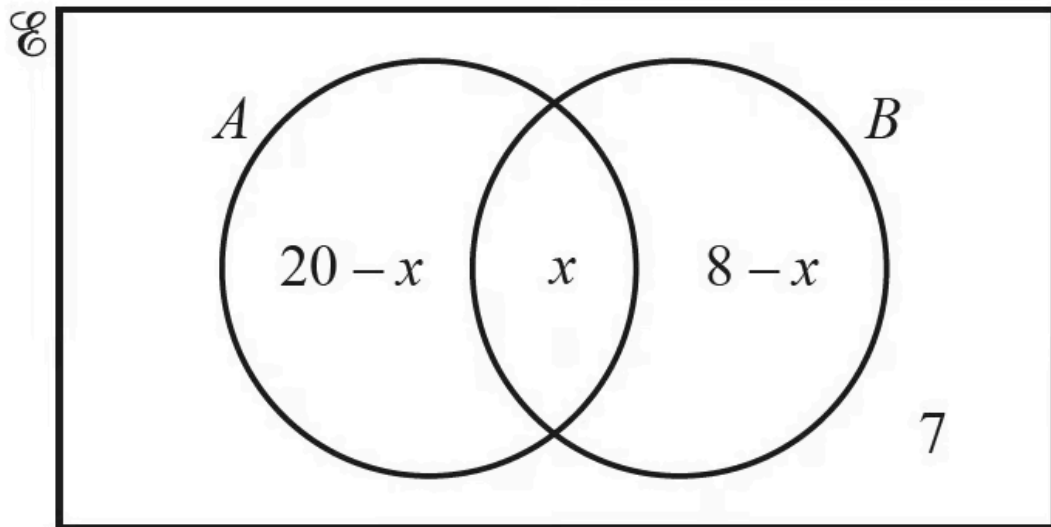
(2 marks)

- 4 In this Venn diagram, shade the region $(A \cup B') \cap C$.



(1 mark)

- 5 The Venn diagram shows information about the number of elements in sets A , B and $\mathcal{E}$.

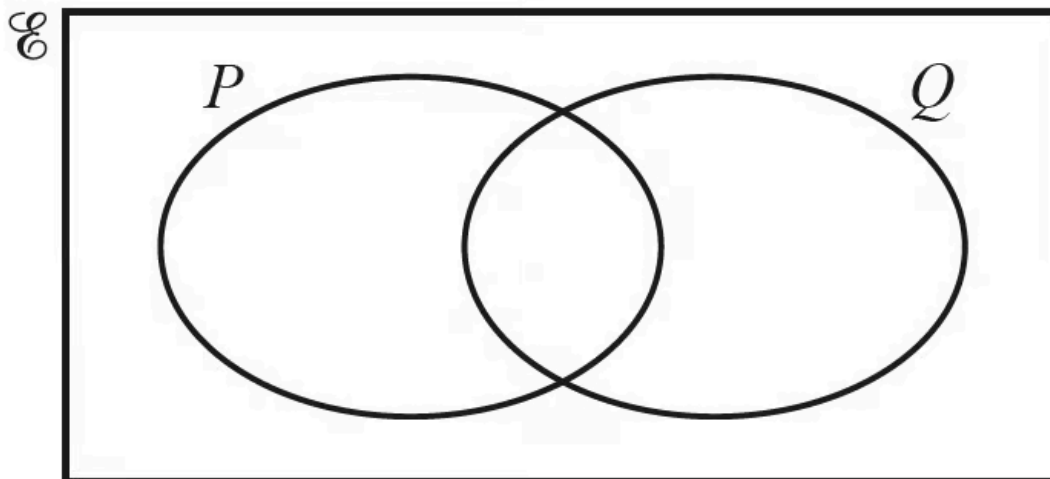


$$n(A \cup B) = 23$$

Find the value of x .

$x = \dots\dots\dots$

(3 marks)



$n(\mathcal{E}) = 20$, $n(P) = 10$, $n(Q) = 13$ and $n(P \cup Q)' = 5$.

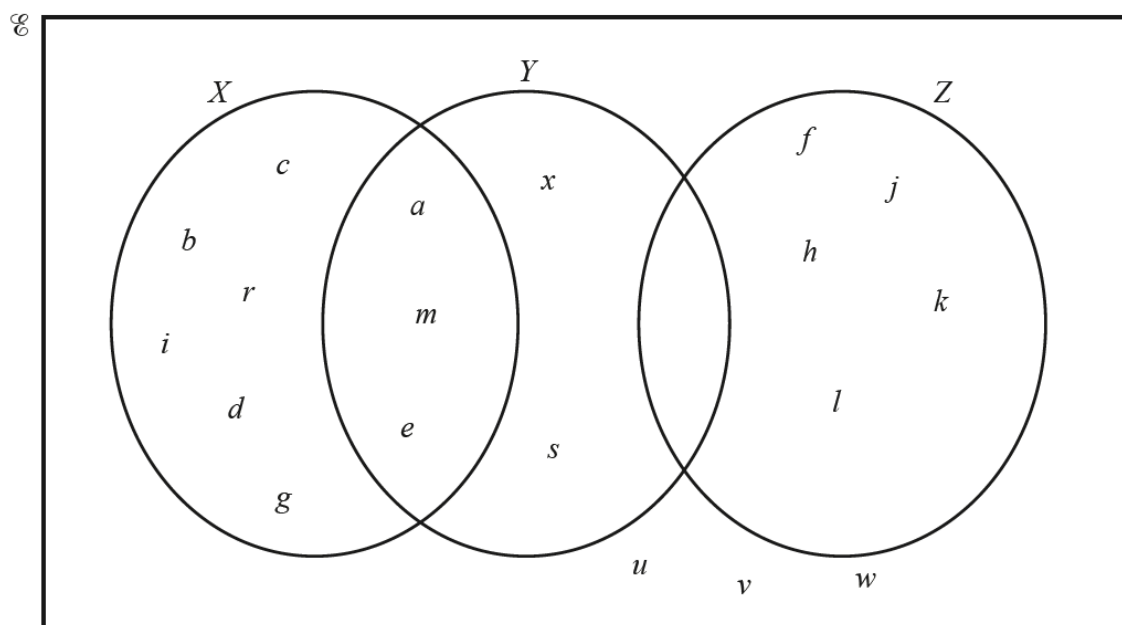
Work out $n(P \cap Q)$.

You may use the Venn diagram to help you.

$n(P \cap Q) = \dots\dots\dots$

(2 marks)

7 (a)



List the elements of $(X \cup Y \cup Z)'$.

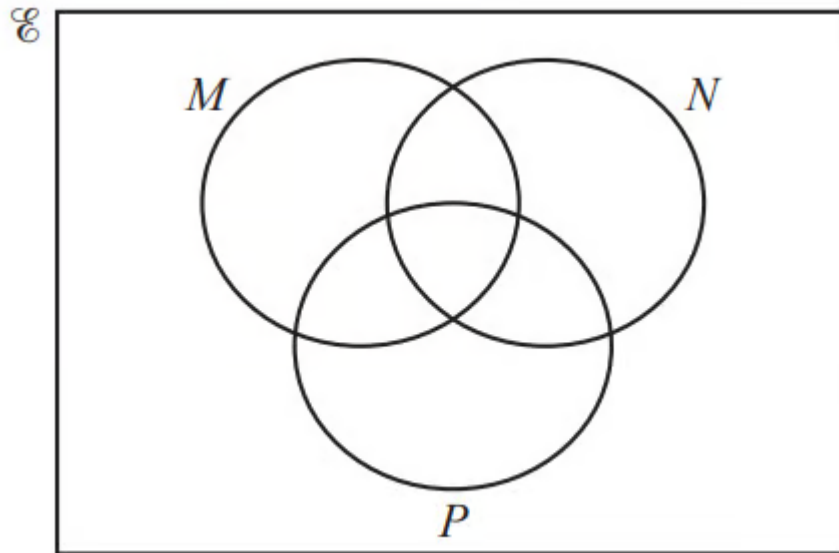
(1 mark)

(b) Find $n(X' \cap Z)$.

(1 mark)

Very Hard Questions

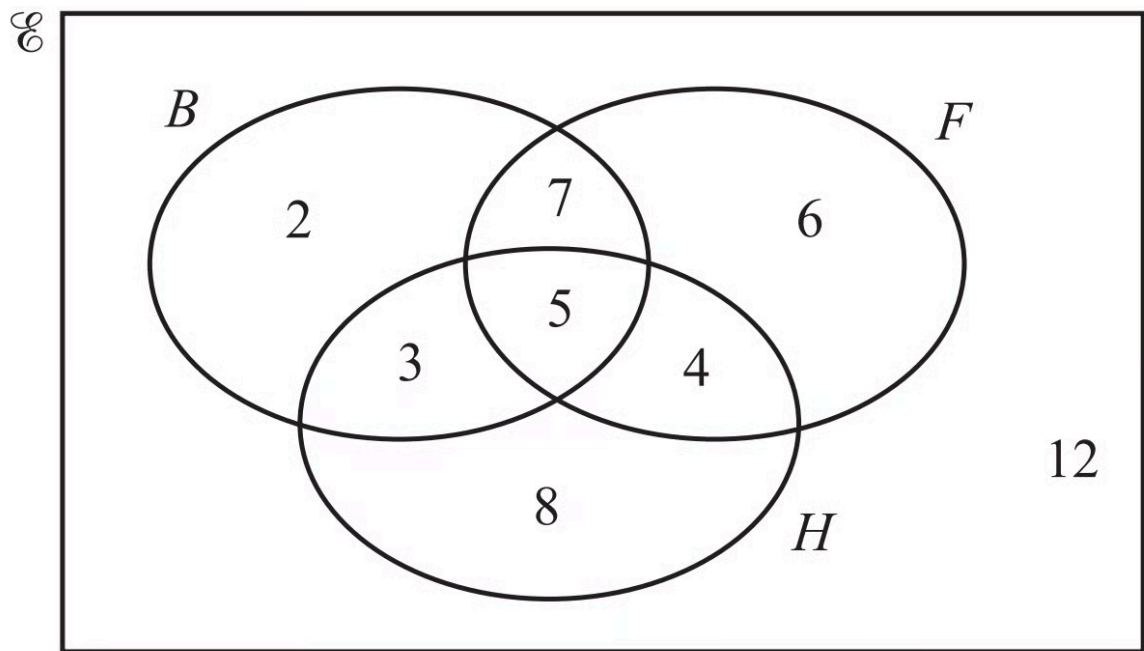
- 1 In this Venn diagram, shade the region $M' \cup N \cup P$.



(1 mark)

- 2 This Venn diagram shows information about the number of students who play basketball

(B), football (F) and hockey (H).

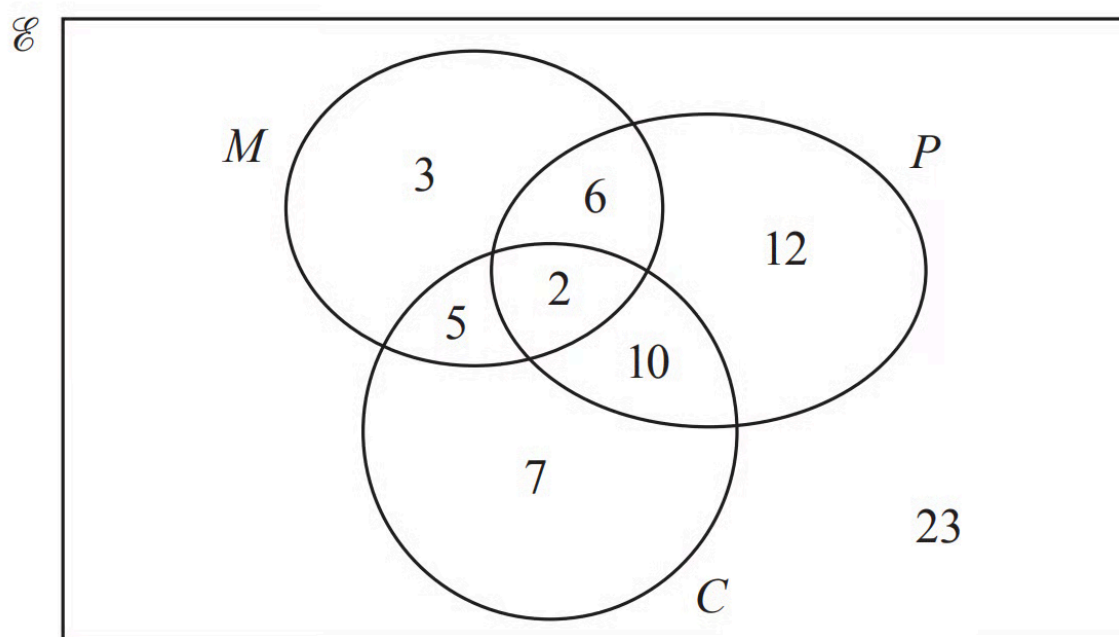


Find $n((B \cup F) \cap H')$.

(1 mark)

3 The Venn diagram below shows information about the number of gardeners who grow

melons (M), potatoes (P) and carrots (C).



Find $n((M \cap P) \cup C')$.

(1 mark)